Work Breakdown Structure – 2-Player Shooting Game Toy	
WBS ID	**Task / Work Package**
1	**Full Game**
1.1	**Project Management**
1.1.1	**Task planning & role distribution**
1.1.2	**Gantt chart creation**
1.2	**User Research & Design Thinking**
1.2.1	**Identify user needs & problems**
1.2.2	**Ideation (mind maps, sketches)**
1.2.3	**Concept selection**
1.3	**Product Design Specification (PDS)**
1.3.1	**Functional requirements**
1.3.2	**Safety requirements**
1.3.3	**Size and constraints**
1.3.4	**Performance criteria**
1.4	**Mechanical Subsystems**
1.4.1	**Shooting Mechanism**
1.4.1.1	Research shooting methods
1.4.1.2	CAD model (Autodesk Inventor)
1.4.1.3	Projectile calculations ★
1.4.1.4	Material selection
1.4.1.5	Manufacturing & assembly
1.4.2	**Vertical Lifting Mechanism**
1.4.2.1	Mechanism research
1.4.2.2	CAD model
1.4.2.3	Load / torque calculations ★
1.4.2.4	Actuator selection & sizing ★
1.4.2.5	Manufacturing & assembly
1.4.3	**Sorting Mechanism**
1.4.3.1	Sorting method selection
1.4.3.2	CAD model ★
1.4.3.3	Sensor selection
1.4.3.4	Manufacturing & assembly
1.5	**Electrical & Electronics System**
1.5.1	**Component selection**
1.5.2	**Circuit design (Proteus)**
1.5.3	**PCB design & fabrication ★**

1.5.4	Wiring diagrams (each subsystem) ★
1.6	Embedded Systems & Software (Game Software)
1.7	System Integration
1.7.1	Mechanical integration
1.7.2	Electrical integration
1.7.3	Wiring and connections
1.8	Testing & Validation
1.8.1	Subsystem testing
1.8.2	Electrical testing
1.8.3	Software testing
1.8.4	Integrated system testing
1.9	Finalization
1.9.1	Final wiring
1.9.2	Final code debugging
1.9.3	Theme & aesthetics completion
1.9.4	Final layout
1.10.	Documentation & Deliverables
1.10.1	Presentation slides
1.10.2	A0 poster
1.10.3	Video demonstration
1.10.4	Bill of Materials (BOM)
1.10.5	Flowcharts & block diagrams ★
1.10.6	GitHub repository organization
1.10.7	WBS ★

COLOUR LEGEND	
Level 1 – Project (dark navy)	**Top-level project node**
Level 2 – Major phases (blue)	e.g. 1.1 Project Management
Level 3 – Sub-phases (mid blue)	e.g. 1.4.1 Shooting Mechanism
Level 4 – Work packages (light blue)	Individual tasks with owners
MILESTONE ITEMS (★)	**Items marked ★ are required for the Week 10 milestone check**
1. Integrated CAD Design	All subsystem CAD models complete
2. Functional Subsystems	Each subsystem independently operational
3. Actuator Sizing	All actuators sized with calculations
4. Projectile Calculations	Range, velocity, force calculations
5. WBS	This document
6. Gantt Chart	14-week schedule
7. PDS	Product Design Specifications
8. System Flowchart	Whole system flowchart
9. System Block Diagram	Whole system block diagram
10. Wiring Diagrams & PCB Schematic	At least one complete PCB schematic

STATUS OPTIONS	**Edit column H on the WBS sheet**
Not started	
In progress	
Complete	
On hold	
At risk	